

IT362: Principles of Data Science

Semester-2, 1447H



Phase 3: Modelling and communication

Topic: Sentiment analysis of reviews of beauty centers

Prepared by

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Phase1:

1. Introduction:

With the widespread use of digital platforms, online reviews have become an essential source for understanding customer experiences and opinions across various services, including beauty centres such as salons and cosmetic clinics. Many customers rely on these reviews before making booking decisions or selecting a specific beauty center.

These reviews often express different sentiments such as satisfaction, dissatisfaction, or neutrality and are usually written as unstructured textual data. Analyzing large volumes of such data manually is time-consuming and inefficient.

This project aims to analyse the sentiments expressed by customers in online reviews of beauty centres using sentiment analysis techniques in order to understand overall customer satisfaction trends and extract insights that support service improvement and decision-making.

Research Question:

What sentiments do customers express in online reviews of beauty centres?

Study	Problem Focus	Dataset	Methods / Models	Key Findings
Study 1 [1]	Difficulty of manually analyzing millions of unstructured reviews	4M Amazon reviews (Kaggle); balanced subset of 500K	ML: LR, SVM, DT, RF, MNB, BNB;DL: LSTM;Features: BoW, TF-IDF, N-grams	TF-IDF with N-grams improved results; LR & SVM reached 91.3%; LSTM achieved highest accuracy (93.3%)
Study 2 [2]	Limited research using advanced NLP for aspect-level insights	2,782 annotated Trustpilot reviews from 5 platforms; 14 aspects	ABSA with POS tagging & bi-terms;Models: LSTM, BERT, DistilBERT, RoBERTa, XLNet	Transformer models outperformed others; shipping & trust were key drivers of sentiment; shipping most influential
Study 3 [3]	Traditional sentiment analysis fails to	Sephora skincare reviews (top 10 products)	Keyword-based aspect detection;VADER + TextBlob;TF-IDF + SMOTE;XGBoost	Strong performance for positive/neutral sentiment; weaker on negative

	capture aspect-level opinions			sentiment; TF-IDF lacks contextual depth
Study 4 [4]	Need to compare ML and DL effectiveness across datasets	Amazon product reviews with star ratings	ML: LR, NB, RF;DL: RNN, CNN	ML models outperformed DL; RF & LR achieved up to 99%; lexicon-based methods performed worst
Study 5 [5]	Lack of aspect-based analysis for a single product across reviews	Wireless earbud reviews from e-commerce platforms	DL & ensembles: LSTM, BERT;Voting+BERT, Bagging+BERT	Ensemble models improved accuracy; Bagging+BERT performed best; BERT strongest base model

2. Literature review:

Common Datasets or Features Used:

Across the reviewed studies, customer reviews serve as the primary data source for sentiment analysis, highlighting their importance in understanding user opinions in e-commerce and service contexts. Several studies relied on Amazon product reviews, while others used reviews from platforms such as Trustpilot and Sephora or focused on specific product categories like wireless earbuds.

In terms of feature representation, traditional machine learning approaches commonly employed Bag-of-Words, TF-IDF, and N-grams, whereas deep learning and transformer-based models relied on contextual embeddings generated by models such as BERT. Additionally, some studies applied aspect-based sentiment analysis to capture sentiment related to specific attributes (e.g., price, quality, shipping), while others focused on overall document-level sentiment classification.

Differences in Modelling Approaches:

The reviewed studies adopt diverse modelling strategies for sentiment analysis. Some focus on traditional machine learning classifiers, such as Logistic Regression, SVM, Random Forest, and Naive Bayes, which emphasize simplicity and interpretability but depend heavily on manual feature engineering.

Other studies employ deep learning models, including LSTM, RNN, and CNN, to better capture sequential and contextual information in text. More recent work shifts toward transformer-based models, such as BERT, DistilBERT, RoBERTa, and XLNet, which provide richer contextual representations and improved performance. In addition, one study applies ensemble learning techniques, combining models such as Voting and Bagging with BERT to enhance robustness and accuracy. Furthermore, some studies extend beyond document-level analysis by using aspect-based sentiment analysis, enabling more detailed insights but often relying on manual annotation or keyword-based extraction, which limits scalability and generalizability.

Strengths and Limitations of Existing Work:

The reviewed studies demonstrate several strengths. Deep learning and transformer-based models, particularly BERT and its variants, consistently outperform traditional approaches by capturing contextual and nuanced sentiment. Aspect-based sentiment analysis provides fine-grained and actionable insights by linking sentiment to specific features, while ensemble learning methods improve model robustness and classification accuracy.

However, notable limitations remain. Traditional machine learning models struggle with contextual understanding, while deep learning and transformer-based methods require large, annotated datasets and significant computational resources, limiting their practical deployment. Many aspect-based approaches depend on manual annotation or keyword-based extraction, reducing automation and generalizability. Additionally, model performance often varies across datasets and domains, making cross-domain transfer challenging.

Missing, Limited, and Unclear Aspects in Previous Work:

Despite progress in sentiment analysis research, several gaps persist. Many studies focus on general sentiment trends rather than detailed aspect-level analysis across

multiple reviews. There is also limited research that systematically compares integrated or ensemble deep learning models under consistent experimental settings. Scalability remains a concern due to reliance on manual annotation and the high computational cost of transformer models. Furthermore, the impact of data preprocessing and model interpretability is often unclear. Additionally, most existing studies concentrate on product-based reviews, with limited attention given to service-oriented domains, where customer experiences are more subjective and context-dependent.

How Our Project Will Build Upon or Differ from Existing Studies:

Our project builds on existing sentiment analysis research by focusing on a service-based domain, specifically beauty centers, rather than the product-focused reviews commonly analyzed in prior work. While many studies centered on e-commerce products, this project emphasizes customer experiences in salons and cosmetic clinics, where feedback is more experience-driven.

Unlike several prior studies that emphasize either large-scale generic sentiment classification or highly complex deep learning and ensemble models, our project adopts a practical and interpretable sentiment analysis approach tailored to service reviews and combines textual sentiment analysis with numerical ratings to examine the relationship between star ratings and expressed sentiment. It also explicitly addresses real-world data challenges, such as short or unclear reviews and potential review bias. By focusing on service quality factors (e.g., cleanliness, pricing, and staff behavior), the project aims to deliver actionable insights that support service improvement and customer satisfaction in the beauty industry.

Overall, this project extends prior sentiment analysis research by applying established

techniques to a new service-oriented application domain, with an emphasis on practical usability and interpretability rather than solely optimizing model performance.

3. Data sources:

Data sources: The dataset consists of unstructured textual reviews (Unstructured Text Data), where each individual review represents one observation (reviews record). Each review is linked to its corresponding beauty centre using a `place_id` collected earlier from the Google Places API. Number of observations: The dataset contains 2,889 reviews collected from 700 beauty salons and spas, where reviews were gathered separately for each place.

Features and Data Types:

- **place_id:** A string identifier used to link each review to its corresponding beauty centre.
- **review_text:** Text data representing the user's written review; this is the primary input for sentiment analysis.
- **rating:** A numeric value ranging from 1 to 5 representing the user's overall rating of the service.
- **time:** A numeric timestamp indicating when the review was published.
- **language:** A categorical variable indicating the language of the review.
- **relative_time_description:** A categorical variable describing the relative posting time (e.g., "one week ago," "one month ago"). The data was integrated at the review level without applying any preprocessing or cleaning steps at this stage, in order to preserve the dataset in its raw form.

The dataset includes several potential sources of bias related to its origin and collection method:

1. **Representation Bias** The data represents only users who chose to write reviews on Google, which may result in unequal representation of certain groups, such as individuals who are less digitally active or who do not typically leave online reviews. In addition, users with extremely positive or negative experiences are more likely to submit reviews, potentially underrepresenting moderate or neutral experiences.
2. **Measurement Bias** The dataset relies on subjective user opinions and ratings, which may be influenced by temporary emotions, isolated incidents, or personal expectations. As a result, the reviews may not consistently reflect the actual or long-term quality of the services provided.
3. **Historical Bias** Some reviews may be outdated and may not reflect the current condition of the beauty centers, especially if there have been changes in management, staff, or service quality. Using such reviews may reinforce perceptions based on past conditions that are no longer valid.

4. Objectives:

Insights:

- Understand overall sentiment trends in online beauty center reviews
- Identify key factors associated with positive and negative sentiments, such as service quality, pricing, cleanliness, and staff behavior

Questions:

Main research question:

What sentiments do customers express in online reviews of beauty centers?

- What are the most common sentiments expressed in beauty center reviews?
- Do positive or negative sentiments dominate customer opinions?
- Is there a relationship between numerical ratings and textual sentiments?

Tasks:

- Classify textual reviews into positive, negative, and neutral sentiments
- Analyze sentiment distribution across a large set of beauty center reviews
- Extract general patterns related to customer satisfaction and dissatisfaction

5. Method:

To achieve the objectives of this project, the following methodology will be applied:

1. Data Collection

Online reviews of beauty centers will be collected from Google Maps using application programming interfaces (APIs).

2. Text Preprocessing

The collected textual data will be prepared by removing irrelevant characters, normalizing text, and handling duplicated or very short reviews to ensure better text quality for analysis.

3. Labeling (Sentiment Annotation)

Each review will be assigned a sentiment label (positive, negative, or neutral) based on semantic meaning.

The labeling process will be supported by numerical ratings, where higher ratings generally indicate positive sentiment and lower ratings indicate negative sentiment, in addition to semantic analysis of the review text. This step enables supervised or semi-supervised sentiment classification.

4. Sentiment Analysis

Sentiment analysis techniques will be applied to classify reviews according to their labeled sentiment categories.

5. Insights Extraction

The classification results will be analyzed to identify overall sentiment trends and to understand key factors influencing customer satisfaction and dissatisfaction.

6. Challenges & Recommendations:

Challenges:

1. Difficulty collecting data from review platforms that rely on dynamic content loading.
2. Presence of reviews written in different languages, which may affect the accuracy of sentiment analysis.
3. Very short or unclear reviews that limit accurate sentiment classification.
4. Review bias, as customers with strong positive or negative experiences are more likely to leave reviews.

Recommendations:

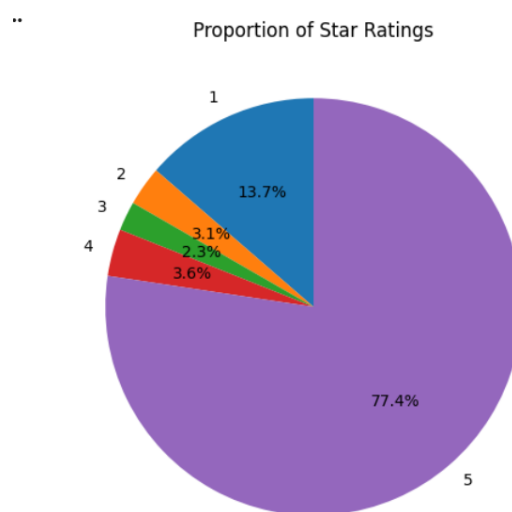
- Use multiple data sources to increase review diversity.
- Apply language detection techniques and process each language appropriately.
- Increase dataset size to reduce the impact of short reviews.
- Consider potential biases when interpreting final results.

Phase2:

Primary Data Exploratory Analysis

This section presents the exploratory data analysis performed on the primary Google beauty reviews dataset to examine rating behavior, textual patterns, relationships between variables, and temporal trends.

Proportion of Star Ratings (Pie Chart)



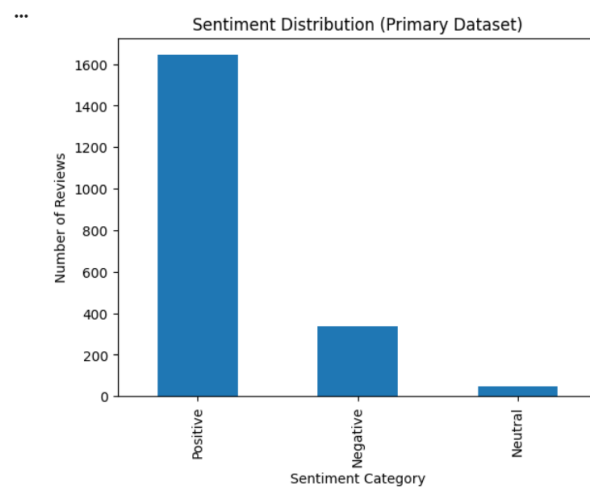
The pie chart illustrates the proportional distribution of star ratings within the primary Google Maps dataset. The results show a clear dominance of 5-star reviews, which account for 77.4% of all entries.

In contrast, 1-star reviews represent 13.7%, while mid-range ratings (2, 3, and 4 stars) collectively constitute a relatively small proportion of the dataset (each below 4%).

This distribution indicates a strong positive sentiment bias, with overwhelmingly favorable customer evaluations. The relatively limited presence of mid-range ratings suggests polarization toward extreme evaluations, particularly toward highly positive feedback.

Overall, the dataset reflects predominantly positive customer perception with comparatively low representation of neutral or moderately rated experiences.

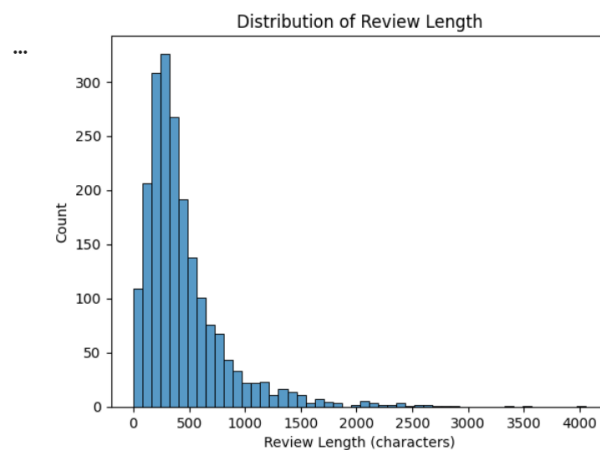
Primary Dataset Sentiment Distribution



Sentiment categories were derived from star ratings and visualized using a bar chart. The results show that positive reviews dominate the dataset, while negative reviews represent a smaller portion, and neutral reviews appear least frequently.

This distribution indicates a strong positive sentiment bias within the dataset, reflecting generally favorable customer experiences.

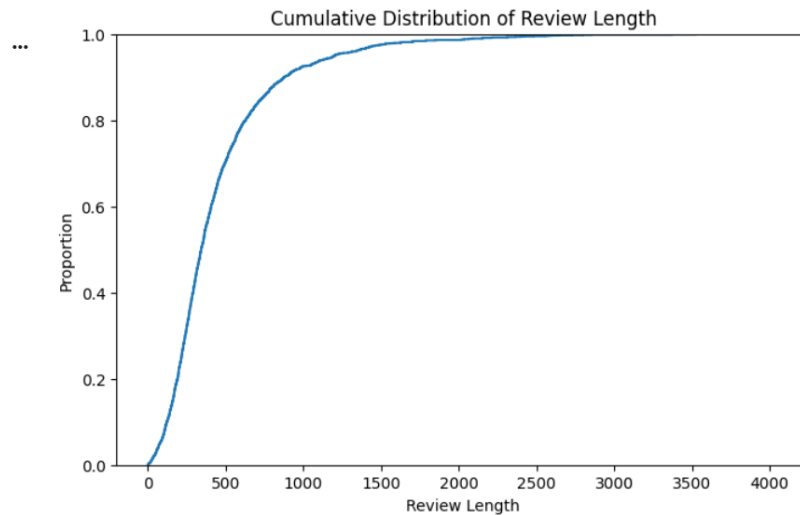
Distribution Review Length



The histogram of review length (in characters) shows a right-skewed distribution. The majority of reviews fall within shorter character ranges, while a small proportion extend to substantially higher character counts.

This pattern suggests that most users provide concise feedback, whereas detailed and lengthy reviews are contributed by a minority of users.

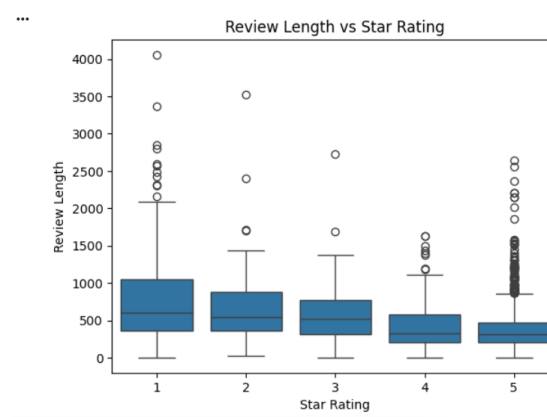
Cumulative Distribution of Review Length (ECDF)



The ECDF plot confirms that a large proportion of reviews are concentrated within lower character lengths. The curve rises steeply in the early range and gradually flattens as review length increases.

This reinforces the observation that long reviews exist but constitute a relatively small fraction of the dataset.

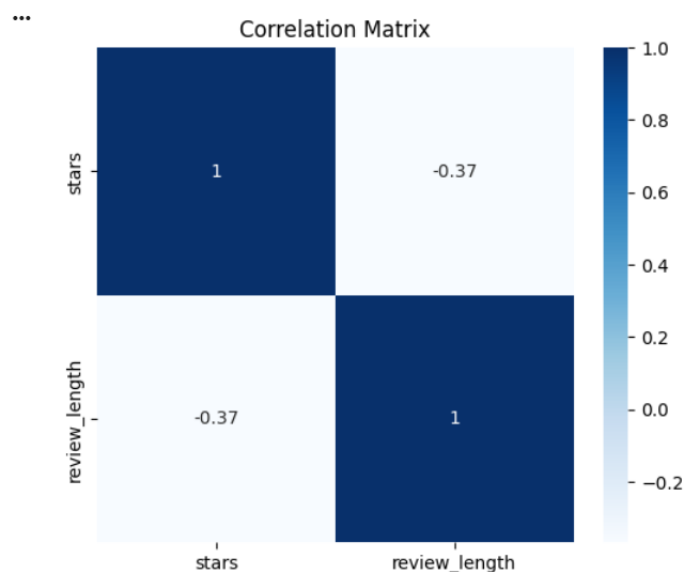
Review Length vs Star Rating



Boxplots comparing review length across star ratings indicate moderate variability within each rating group. Lower star ratings appear to show slightly higher variability and occasional extreme outliers.

However, the central tendencies across rating groups do not demonstrate a strong structural difference. This suggests that review length alone is not a dominant determinant of rating level.

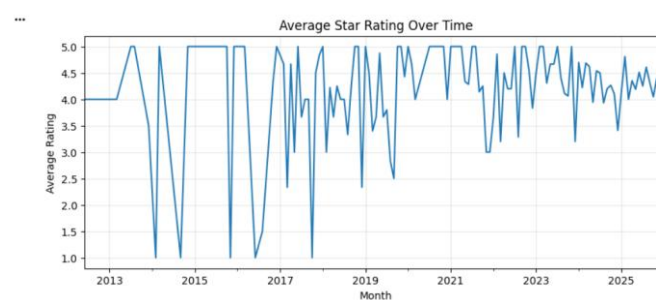
Correlation Analysis



The correlation matrix indicates a moderate negative correlation (-0.37) between star rating and review length.

This suggests that lower-rated reviews tend to be somewhat longer, potentially reflecting more detailed explanations or complaints. However, the magnitude of this relationship indicates that rating level alone does not strongly determine textual length.

Trend Analysis



The time-series analysis of average star ratings over the observed period shows noticeable short-term fluctuations. However, the overall pattern remains relatively stable, with no clear sustained upward or downward trend.

This indicates consistent customer sentiment over time without structural volatility.

Text-Based Exploratory Analysis

- Top 20 Most Frequent Words in the Primary Dataset

0	experience	718
1	time	700
2	treatment	616
3	recommend	595
4	feel	583
5	staff	582
6	service	578
7	hair	574
8	would	546
9	really	522
10	great	509
11	amazing	486
12	massage	474
13	back	459
14	always	442
15	friendly	417
16	professional	401
17	lovely	396
18	made	383

This figure presents the 20 most frequently occurring terms in the processed review texts after applying text preprocessing techniques, including tokenization, stopword removal, and lemmatization. The results were generated using CountVectorizer to calculate overall term frequency across the corpus.

The dominant terms such as experience, time, treatment, recommend, staff, and service indicate that customers primarily focus on service quality, staff interaction, and overall experience. Additionally, positive descriptive words such as great and amazing appear frequently, suggesting a generally positive sentiment within the primary dataset.

These findings provide an initial understanding of the main discussion themes before moving to more contextual textual analysis.

-Top 10 Bigram

	bigram	count
0	highly recommend	253
1	made feel	167
2	customer service	108
3	first time	106
4	feel comfortable	89
5	make feel	83
6	took time	78
7	would recommend	72
8	would highly	71
9	highly recommended	68

This figure illustrates the most frequent two-word combinations (bigrams) extracted using CountVectorizer with `ngram_range=(2,2)`. Unlike single-word frequency analysis, bigrams capture contextual relationships between terms, allowing for more meaningful interpretation of customer feedback.

Frequent bigrams such as highly recommend, customer service, first time, and feel comfortable indicate strong satisfaction signals and recurring service-related themes. The presence of recommendation-related phrases suggests high levels of trust and positive customer endorsement.

This analysis enhances interpretability by revealing patterns that single-word frequencies alone may not capture.

-TF-IDF Analysis

1	great	0.025645
2	service	0.025540
3	hair	0.025483
4	experience	0.025350
5	recommend	0.024798
6	time	0.024693
7	amazing	0.024100
8	treatment	0.024043
9	staff	0.023689
10	feel	0.022361
11	really	0.022285
12	good	0.021703
13	friendly	0.021425
14	always	0.021292
15	lovely	0.020952
16	would	0.020885
17	back	0.020611
18	nail	0.020320
19	professional	0.019945

This figure presents the most informative and distinctive terms identified using TF-IDF vectorization. Unlike simple frequency counts, TF-IDF assigns higher importance to terms that are significant within specific reviews but not overly common across the entire corpus.

Terms such as massage, treatment, professional, and nail appear with higher TF-IDF scores, indicating that these words carry stronger discriminative value within the dataset. This suggests that certain service categories or specialized treatments are particularly emphasized in customer discussions.

TF-IDF analysis helps highlight meaningful patterns beyond commonly repeated words and supports deeper textual understanding.

Summary of Findings

The exploratory data analysis of the primary Google Maps beauty reviews dataset reveals several consistent structural and behavioral patterns. The rating distribution is

heavily concentrated toward 5-star reviews, which account for 77.4% of all entries, indicating a strong positive sentiment bias.

Review length follows a right-skewed distribution, where the majority of reviews are relatively short, while a small subset extends to substantially higher character counts.

A moderate negative correlation (-0.37) was observed between star rating and review length, suggesting that lower-rated reviews tend to be slightly longer and potentially more detailed.

Time-series analysis indicates that average star ratings remain relatively stable over time, with short-term fluctuations but no sustained upward or downward trend.

Overall, the dataset reflects predominantly positive customer sentiment with consistent temporal behavior and moderate variability in textual depth.

Key Findings

- 5-star reviews dominate the dataset, representing 77.4% of all entries.
- 1-star reviews account for 13.7%, while mid-range ratings (2–4 stars) each represent less than 4% of the dataset.
- Review length exhibits a right-skewed distribution with a long tail of extreme values.
- A moderate negative correlation (-0.37) exists between star rating and review length.
- Lower-rated reviews tend to be slightly longer, suggesting more detailed feedback in negative experiences.
- Average rating trends remain relatively stable over time without a clear long-term directional shift.

Limitations

The dataset was collected from Google Maps and represents user-generated content, which may introduce subjective bias, exaggerated opinions, or inconsistent writing styles.

The strong concentration of positive ratings (77.4%) creates class imbalance, which may influence interpretation of sentiment-related patterns and limit the representativeness of mid-range experiences.

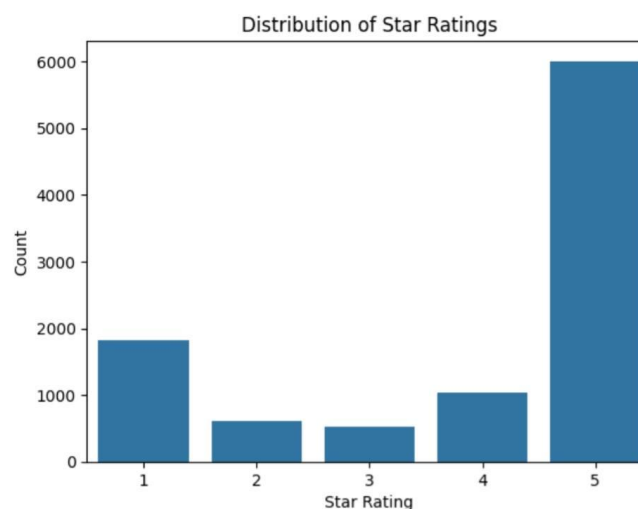
Additionally, correlation analysis identifies statistical associations but does not establish causation. Therefore, the observed relationship between review length and star rating (-0.37) should be interpreted cautiously.

Finally, contextual variables such as demographic information, service type variations, or external market factors are not fully captured in the dataset, which may limit deeper causal interpretation of customer sentiment patterns.

Secondary Data Exploratory Analysis

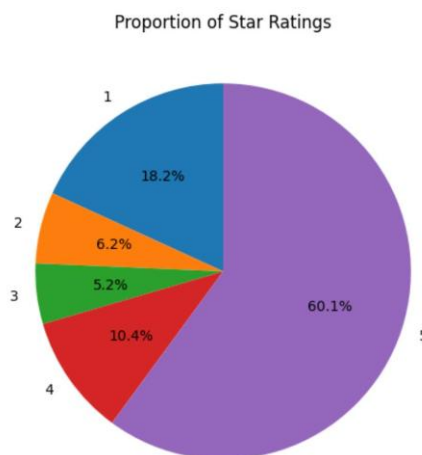
This section presents the exploratory data analysis conducted on the secondary Yelp beauty reviews dataset to identify patterns, relationships, and temporal trends.

Distribution of Star Ratings



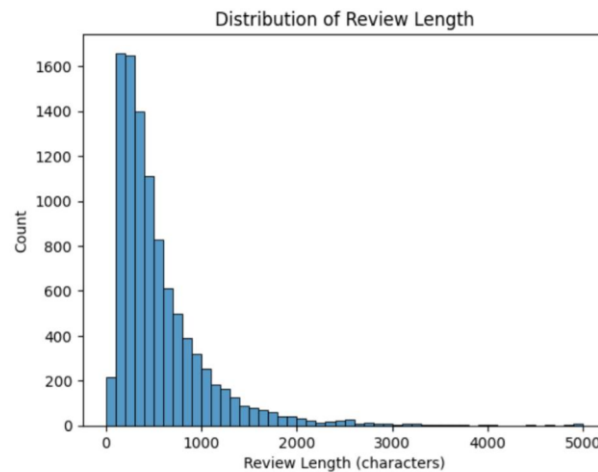
It presents the distribution of star ratings using both a bar chart and a pie chart. The results show that 5-star ratings dominate the dataset, accounting for approximately 60% of all reviews. One-star reviews represent about 18%, while 2- and 3-star ratings are less frequent. This indicates a generally positive sentiment trend with slight polarization toward extreme ratings.

Proportion of Star Ratings (Pie Chart)



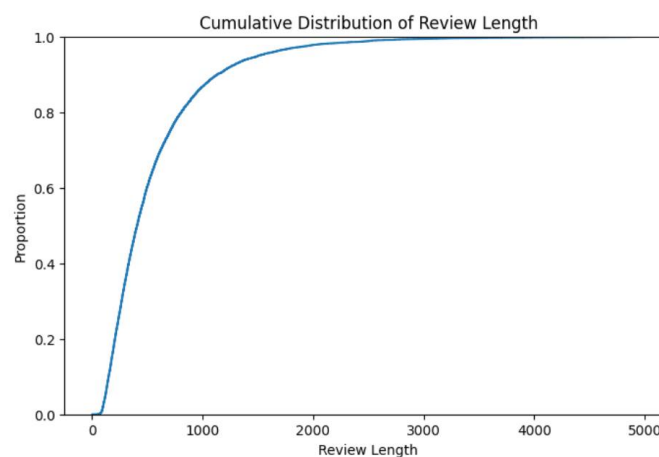
It presents the proportional distribution of star ratings using a pie chart. Consistent with the bar chart results, 5-star reviews account for approximately 60% of all entries. One-star reviews represent about 18%, while mid-range ratings (2–3 stars) appear less frequently. The pie chart visually emphasizes the dominance of positive reviews within the dataset.

Distribution of Review Length



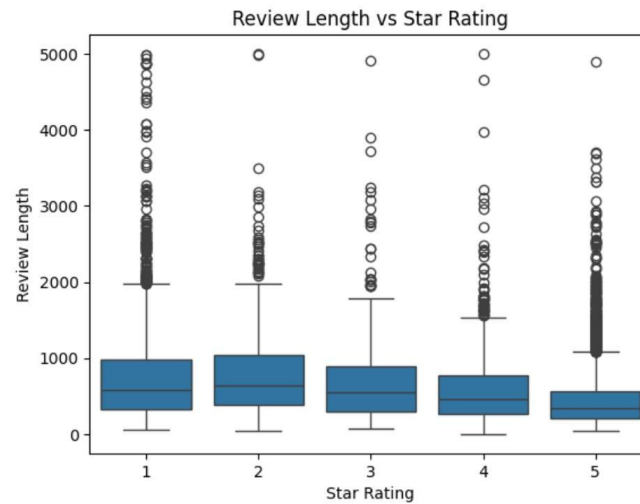
The histogram of review length (in characters). The distribution is strongly right-skewed, with the majority of reviews falling under 1,000 characters. However, a long tail extends beyond 3,000 characters, reaching over 5,000 characters in extreme cases.

Cumulative Distribution of Review Length (ECDF)



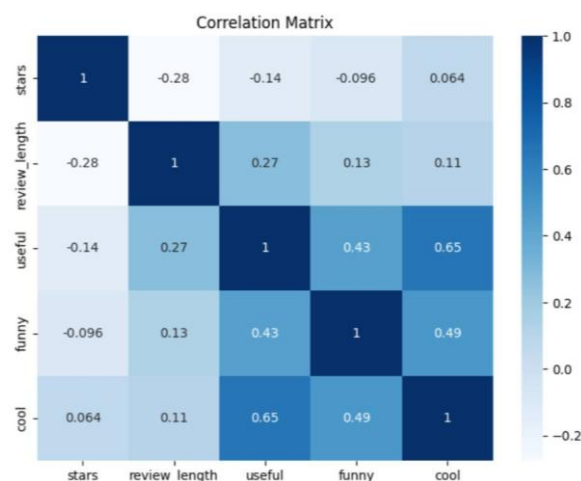
(ECDF) confirms that most reviews are concentrated within shorter lengths, while only a small proportion exceed high character counts. This suggests that while most users provide brief feedback, a minority contribute detailed reviews.

Review Length vs Star Rating



It presents boxplots comparing review length across star ratings. Lower ratings (particularly 1 and 2 stars) show slightly higher median review lengths compared to higher ratings. However, the differences are not substantial, indicating no strong dependency between rating level and review length. Several extreme outliers are observed across all rating groups.

Correlation Analysis



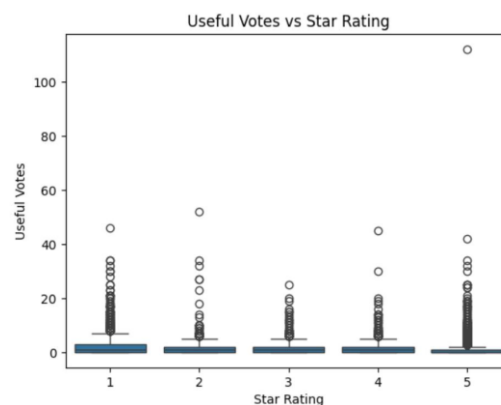
It presents the correlation matrix among numerical variables. A moderate negative correlation (-0.28) is observed between star rating and review length, suggesting that lower ratings tend to be slightly longer.

Review length shows a small positive correlation (0.27) with useful votes.

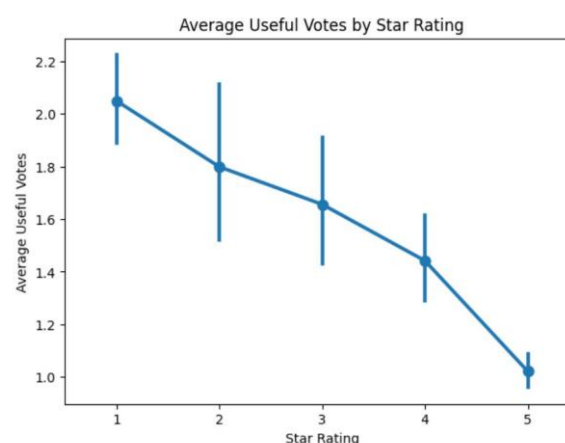
Engagement metrics (useful, funny, cool) demonstrate moderate to strong positive correlations among themselves, with the strongest observed between useful and cool (0.65).

Weak negative correlations were observed between star ratings and engagement metrics.

Engagement vs Star Rating



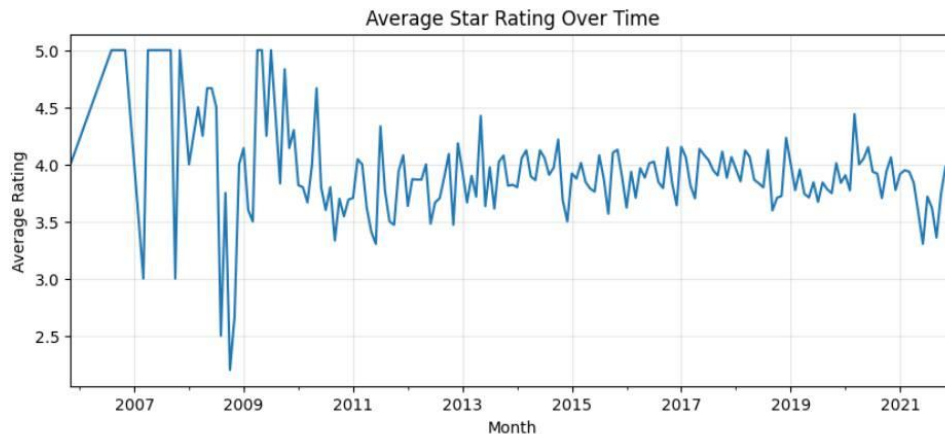
The distribution of useful votes across star ratings. Most reviews receive low useful counts regardless of rating level, although several extreme outliers are observed, particularly among 5-star reviews. This indicates that engagement levels vary substantially within each rating category.



It presents the average useful votes by star rating. The results show a decreasing trend in average useful votes as star ratings increase. Lower-rated reviews tend to receive slightly higher average useful votes. This may suggest that critical reviews provide

more detailed or informative feedback. However, the relationship appears moderate rather than strongly linear, and star rating alone does not strongly determine engagement levels.

Trend Analysis

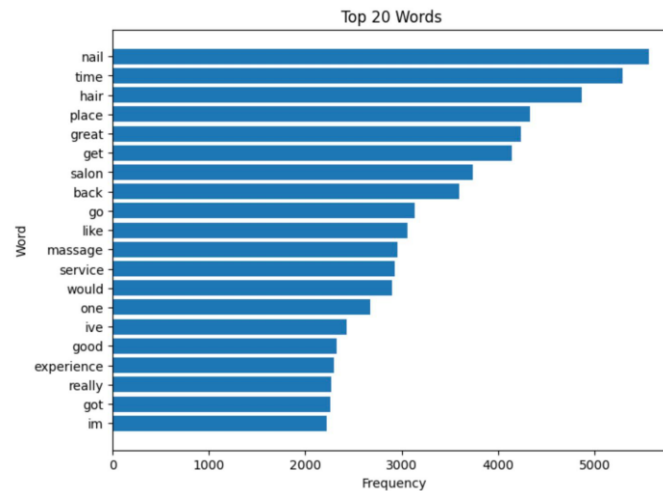


The average star rating over time. Early years show noticeable fluctuations, but ratings gradually stabilize between approximately 3.7 and 4.1 stars. No clear long-term upward or downward trend is observed, indicating relatively stable customer sentiment over time.

Text-Based Exploratory Analysis

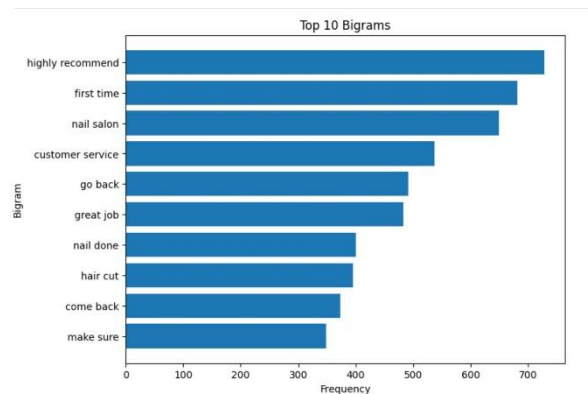
In addition to the numerical and relational analyses, text-based exploratory analysis was conducted to examine the linguistic patterns and dominant themes within customer reviews.

Top 20 Most Frequent Words



It presents the most frequently occurring words in the review text. The dominant terms mainly relate to beauty services (e.g., nail, hair, salon) along with positive expressions such as great and good, reflecting generally positive customer experiences.

Top 10 Bigrams



It presents the most frequent two-word combinations (bigrams). Expressions such as “highly recommend,” “customer service,” and “nail salon” indicate strong service-related themes and positive customer satisfaction

TF-IDF Analysis

Out[33]:

	term	avg_tfidf
0	nail	0.040427
1	hair	0.033475
2	great	0.033060
3	place	0.030716
4	time	0.029759
5	salon	0.027029
6	massage	0.024220
7	get	0.024210
8	back	0.023764
9	service	0.022265
10	go	0.021545
11	cut	0.019795
12	love	0.019516
13	ive	0.019374
14	always	0.019153
15	like	0.018976
16	job	0.018815
17	good	0.018596
18	would	0.018350
19	best	0.018109

TF-IDF analysis highlights the most representative and distinctive terms in the dataset. Service-related keywords such as nail, hair, and salon receive higher importance scores, confirming the consistency of dominant beauty-service themes.

Summary of Findings

The exploratory data analysis of the secondary Yelp beauty reviews dataset reveals several key patterns. A dominant proportion of reviews are 5-star ratings, indicating generally positive customer sentiment. Review length follows a right-skewed distribution, where most reviews are relatively short, while a small number extend to extreme lengths.

A moderate negative correlation (-0.28) was observed between star rating and review length, suggesting that lower-rated reviews tend to be slightly longer. Engagement metrics (useful, funny, cool) demonstrate moderate to strong positive correlations among themselves, while only weak relationships were observed between star ratings and engagement levels.

Time-series analysis shows that average ratings remain relatively stable over time, without a clear long-term upward or downward trend.

Text-based exploratory analysis further reinforces these findings. Word frequency,

bigram analysis, and TF-IDF results reveal dominant beauty-service themes and positive expressions that align with the rating distribution patterns. Overall, the dataset reflects stable and generally positive customer sentiment, with engagement behavior influenced more by review content characteristics than by star rating alone.

Key Findings

- 5-star reviews dominate the dataset, representing approximately 60% of all entries.
- Review length exhibits a right-skewed distribution with a long tail of extreme values.
- A moderate negative correlation (-0.28) exists between star rating and review length.
- Review length shows a small positive correlation (0.27) with useful votes.
- Engagement metrics demonstrate moderate to strong positive interrelationships.
- Star rating shows only weak associations with engagement behavior.
- Average rating trends remain stable over time.
- Text-based analysis identifies dominant beauty-service themes and positive customer expressions consistent with rating patterns.

Limitations

The dataset originates from a secondary source and was not collected specifically for this study. Therefore, potential sampling biases or inconsistencies in data collection methods may affect the generalizability of the findings.

User-generated reviews may contain subjective bias, exaggerated opinions, or inconsistencies in writing style. Additionally, contextual factors such as location, service type, or demographic information are not fully captured in the dataset, which may limit deeper interpretation of sentiment patterns.

Correlation analysis identifies associations but does not establish causation.

Therefore, observed relationships between variables should be interpreted cautiously.

Furthermore, text-based exploratory techniques such as word frequency, bigram analysis, and TF-IDF focus primarily on surface-level linguistic patterns. These

methods may not fully capture contextual meaning, sarcasm, implicit sentiment, or complex semantic relationships within the review text.

Comparison

This section presents a comparison between the primary and secondary datasets. Key statistical metrics and overall trends are examined to identify similarities and differences between the two data sources.

Compare Key Metrics

Statistical Comparison

To compare the primary (Google Reviews) and secondary (Yelp Reviews) datasets, summary statistics and trend analysis were conducted using the common variable “stars” (rating).

The summary statistics reveal that the primary dataset has a higher average rating (Mean = 4.35) compared to the secondary dataset (Mean = 3.88). Despite this difference in mean values, both datasets share the same median value of 5, indicating that high ratings dominate in both sources. However, the secondary dataset exhibits a higher standard deviation, suggesting greater variability in user ratings compared to the primary dataset. This indicates that Yelp reviews are more dispersed and less consistently positive than Google reviews.

In addition to summary statistics, a time-based trend analysis was performed by calculating the monthly average star rating. In the primary dataset, the average rating remains relatively stable over time, with only minor short-term fluctuations and no clear long-term upward or downward trend. Similarly, the secondary dataset shows noticeable fluctuations in earlier periods but eventually stabilizes around a relatively consistent level (mostly between approximately 3.7 and 4.1 in most months), with occasional deviations. No sustained long-term increase or decline is observed in either dataset.

Overall, the comparison of key metrics indicates that while both datasets demonstrate generally positive rating behavior, the primary dataset is more concentrated at higher

rating levels, whereas the secondary dataset reflects greater variability in user evaluations.

Metric Comparison Table

Metric	Primary Dataset (Google Reviews)	Secondary Dataset (Yelp Reviews)
Number of Reviews (count)	2894	10000
Mean Rating	4.354181	3.880700
Median Rating	5	5
Standard Deviation (Std)	1.382569	1.587488
First Quartile (Q1 – 25%)	5	3
Third Quartile (Q3 – 75%)	5	5

Contextualize Findings

Pattern Validation

The secondary dataset was used to provide broader context and validate the patterns observed in the primary dataset. First, both datasets demonstrate generally positive rating behavior, as high ratings dominate in each source. While the primary dataset shows a higher average rating, the overall tendency toward high ratings is still consistent across both platforms, supported by the shared median value of 5.

In addition, trend analysis of monthly average star ratings indicates that customer satisfaction remains relatively stable over time in both datasets. The primary dataset shows minor short-term fluctuations without a sustained upward or downward trend, and the secondary dataset displays early variability followed by stabilization within a consistent range, also without a long-term directional change. This suggests that satisfaction patterns are not driven by temporary spikes but reflect steady user evaluations over time.

Finally, both datasets reveal a negative relationship between star rating and review length, indicating that lower-rated reviews tend to be longer and more detailed, while higher-rated reviews are often shorter. The presence of this pattern in both the primary and secondary data sources strengthens the validity of the primary findings and suggests that these behaviors are consistent across different review platforms within the beauty-services domain

Explaining Differences

Although both datasets demonstrate generally positive rating behavior, a clear difference exists in their average rating levels. The primary dataset (Google Reviews) exhibits a higher mean rating compared to the secondary dataset (Yelp Reviews).

This discrepancy may be explained by differences in platform characteristics and user behavior. Google Reviews are often submitted quickly and may reflect brief, convenience-based feedback, which can lead to more generous ratings. In contrast, Yelp users may be more detail-oriented and critical in their evaluations, resulting in a wider distribution of ratings.

Additionally, differences in sample size may influence the observed variation. The secondary dataset contains a larger number of reviews, which may capture a broader range of customer experiences, including more neutral or negative feedback.

Overall, while both datasets indicate positive customer satisfaction trends, structural and behavioral differences between platforms likely account for the variation in average ratings.

Summary

The comparative analysis suggests that although both datasets reflect generally positive customer satisfaction in beauty-related services, platform-specific patterns are evident. The primary dataset consistently shows higher ratings, while the secondary dataset demonstrates greater variability in user evaluations. This indicates that user rating behavior may not only reflect service quality but also platform-related dynamics.

The analysis also highlights that rating trends remain stable over time across both datasets, suggesting that customer perception of beauty services has remained relatively consistent throughout the observed period.

Based on these findings, several hypotheses can be proposed. First, platform design and review culture may influence how users assign ratings, potentially leading to systematically higher ratings on Google compared to Yelp. Second, Yelp users may exhibit more critical or detailed evaluation behavior, contributing to greater variability in ratings. Third, differences in user demographics or regional distribution may partially explain the variation between the two datasets.

These insights open opportunities for further investigation into how digital platforms shape customer feedback behavior.

Phase3:

Data Preparation for Modelling

In this phase, the cleaned dataset obtained from Phase 2 was prepared for machine learning modelling. The dataset consists of processed textual reviews and their corresponding sentiment labels.

The 'processed_text' column was selected as the input feature, while the 'sentiment_label' column was used as the target variable. Any missing values were removed to ensure data quality and avoid issues during model training.

To enable machine learning models to process textual data, two feature extraction techniques were applied. First, the Bag of Words (BoW) method was used to represent text as word frequency vectors. Second, the TF-IDF technique was applied to assign importance weights to words based on their frequency across the dataset, which helps reduce the influence of common words.

The dataset was then split into training and testing sets using an 80/20 ratio. A fixed random state was used to ensure reproducibility. Additionally, stratified sampling was applied to preserve the distribution of sentiment classes in both sets, ensuring a fair evaluation of the models.

Finally, validation checks were conducted to confirm that there were no missing values, the data types were correct, and the class distribution remained consistent after splitting. The dataset is now fully prepared and ready for model training and evaluate.

Baseline Model

To establish a reference point for model performance, a baseline classification model was implemented using Logistic Regression. The processed review text was transformed into numerical features using TF-IDF vectorization, and the dataset was split into training and testing sets.

The model was evaluated using accuracy, precision, recall, and F1-score. The results indicate that the baseline model achieved an accuracy of 89.3% and an F1-score of 87.3%, demonstrating strong overall performance.

However, class-wise analysis revealed that the model performs significantly better on the positive class compared to the neutral and negative classes. In particular, the neutral class was poorly predicted, likely due to class imbalance and the relatively small number of neutral samples in the dataset.

These results establish a solid benchmark for comparison with more advanced models.

Advanced Model 1: Support Vector Machine (SVM)

To further enhance the performance of the sentiment classification task, a Support Vector Machine (SVM) model was implemented. SVM is a powerful classification algorithm that is well-suited for high-dimensional data such as TF-IDF text representations.

The model was trained using the same TF-IDF features and the same training and testing split used in previous models to ensure a fair comparison. A LinearSVC classifier was applied to learn the relationship between textual features and sentiment labels.

The model was evaluated using accuracy, precision, recall, and F1-score. The results show that the SVM model achieved an accuracy of approximately 94.3%, indicating strong overall performance.

A detailed class-wise analysis shows that the model performs very well on the

positive class, achieving high recall and precision. The negative class also shows good performance with some minor misclassification. However, similar to previous models, the neutral class was poorly predicted due to class imbalance and the limited number of neutral samples in the dataset.

Overall, the SVM model demonstrates strong performance and effective generalization, making it a competitive model compared to both the baseline and Random Forest models.

Advanced Model 2: Random Forest

To further improve the performance of the sentiment classification task, an advanced model was implemented using the Random Forest algorithm. Random Forest is an ensemble learning method that combines multiple decision trees to produce more accurate and stable predictions compared to single-model approaches.

The model was trained using the same TF-IDF feature representation applied in the baseline model to ensure a fair comparison. The dataset was also split into training and testing sets using the same configuration. A Random Forest classifier with 100 decision trees was used, allowing the model to capture more complex patterns in the data.

The model was evaluated using accuracy, precision, recall, and F1-score. The results show that the Random Forest model achieved an accuracy of 91.9% and an F1-score of 90.4%, indicating an improvement over the baseline model.

A detailed analysis of class-wise performance shows that the model performs very well on the positive class, achieving high recall and precision. The negative class shows moderate performance, with some misclassification still present. However, similar to the baseline model, the neutral class was poorly predicted. This is likely due to class imbalance, as the number of neutral samples is significantly lower than the other classes.

Overall, the Random Forest model demonstrates stronger performance and better generalization compared to the baseline model. These results make it a strong candidate for selection as the best-performing model in the final evaluation stage.

Model Evaluation and Selection

In this phase, the performance of all developed models was evaluated and compared in order to determine the best-performing model for sentiment prediction. Three models were considered: Logistic Regression as the baseline model, and Support Vector Machine (SVM) and Random Forest as advanced models.

All models were trained and tested using the same dataset, feature representation (TF-IDF), and data split configuration to ensure a fair comparison. The evaluation was conducted using four main performance metrics: Accuracy, Precision, Recall, and F1-score.

The results show that the SVM (LinearSVC) model achieved the highest performance among all models, with an accuracy of approximately 94.3% and an F1-score of 93.48%. In comparison, both Logistic Regression and Random Forest achieved lower performance, with accuracy values around 91.88% and F1-scores of approximately 90.41%.

These results indicate that the SVM model is more effective in capturing patterns within textual data, especially when using TF-IDF features. Additionally, the F1-score was considered the most important metric in this comparison, as it provides a balance between precision and recall, making it more reliable for classification tasks.

To further analyze the performance of the best model, a confusion matrix was generated for the SVM model. The results show that the model performs very well in predicting the positive class, with a high number of correct classifications. The negative class also shows strong performance with only minor misclassifications. However, the neutral class was the most challenging to predict, as it exhibited a higher rate of misclassification.

This limitation is likely due to class imbalance in the dataset, where the neutral class contains fewer samples.

Model Application on Unseen Reviews

After selecting Support Vector Machine (SVM) as the best-performing model based on accuracy, precision, recall, and F1-score, the model was applied to new, unseen customer reviews that were not part of the training or testing datasets.

The same preprocessing and TF-IDF feature transformation pipeline was used to ensure consistency. The model successfully classified clearly positive and negative reviews, demonstrating its ability to generalize to real-world inputs.

However, the model showed limited capability in identifying neutral sentiment. This behavior is consistent with earlier evaluation results, where the neutral class had very low recall and precision. The likely cause of this limitation is class imbalance, as the dataset contains significantly fewer neutral samples compared to positive and negative reviews.

Overall, this step confirms that the selected model can be used for practical sentiment classification, particularly for distinguishing strongly positive and negative customer opinions.

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